AP & RAP Assignment 4

1. Suppose that $(\Omega, \mathcal{A}, \mu)$ is a measure space and a measurable function f is bounded μ-a.e. and that $\mu(\Omega) < \infty$. Prove or disprove that f is μ-integrable.

2. Let f_1, f_2, $\cdots$ be measurable functions on $(\Omega, \mathcal{A}, \mu)$. Prove that if $\sum_{n=1}^{\infty} \int_\Omega |f_n| d\mu < \infty$, then $\sum_{n=1}^{\infty} |f_n| < \infty$ holds almost everywhere with respect to $[\mu]$ and $\int_\Omega (\sum_{n=1}^{\infty} f_n) d\mu = \sum_{n=1}^{\infty} \int_\Omega f_n d\mu$.

3. Let $(\Omega, \mathcal{A}, \mu)$ be a measure space and $\{A_n\}_{n\in\mathbb{N}}$ be a sequence of sets in $\mathcal{A}$. Define

$$\liminf_{n\to\infty} A_n := \bigcup_{n=1}^{\infty} \bigcap_{m=n}^{\infty} A_m, \qquad \limsup_{n\to\infty} A_n := \bigcap_{n=1}^{\infty} \bigcup_{m=n}^{\infty} A_m.$$

   Prove that:

$$\mu(\liminf_{n\to\infty} A_n) \le \liminf_{n\to\infty} \mu(A_n).$$

   If μ is a finite measure, then also

$$\limsup_{n\to\infty} \mu(A_n) \le \mu(\limsup_{n\to\infty} A_n).$$

4. Suppose that $(\Omega, \mathcal{A}, \mu)$ is a measure space and f is integrable and satisfies $f > 0$ μ-a.e. Show that $\int_A f d\mu > 0$ for every $A \in \mathcal{A}$ such that $\mu(A) > 0$.

5. Let μ be a σ-finite measure on the measurable space $(\Omega, \mathcal{A})$. If f and g are measurable, $\int_\Omega f d\mu$ and $\int_\Omega g d\mu$ exist, and $\int_A f d\mu \le \int_A g d\mu$ for any $A \in \mathcal{A}$, then it follows that $f \le g$ almost everywhere.

6. Let $(\Omega, \mathcal{A}, \mu)$ be a measure space and $f \in \mathcal{L}^1(\mu)$. Prove that for any $\varepsilon > 0$ there exists $A \in \mathcal{A}$ such that $\mu(A) < \infty$ and $|\int_A f d\mu - \int_\Omega f d\mu| < \varepsilon$.